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Data Science FINAL PROJECT REPORT

Project Title:

Predicting Crop Yield Using Machine Learning: A Data-Driven Approach

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

I did not use human participants in my MSc Project.

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SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

Abstract

The current situation of global food security faces two main dangers which are climate change and the depletion of vital resources. Sustainable precision farming depends on accurate agricultural output forecasting which serves as its fundamental requirement. My study addresses this urgent need through the development of a high-precision machine learning framework which predicts crop yields based on environmental and agricultural factors. I used the Agriculture Crop Yield dataset to test the performance of Linear Regression, Decision Tree, and Random Forest models by measuring their results with MAE, MSE, and R^2 metrics. The research found that the optimized Random Forest model reached an R^2 score of 0.913 while maintaining an RMSE value of 0.50. I demonstrated through SHAP explainable AI that rainfall and fertilizer application together with irrigation function as the main factors which determine productivity. This study demonstrates that my predictive system functions as a reliable and clear tool which helps farmers to allocate resources effectively within their agricultural operations..

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Chapter 1: Introduction

Overview

Accurate crop yield prediction is essential for improving agricultural management and ensuring food security. The study of environmental data and soil properties together with farming practices for crop yield prediction depends on machine learning and remote sensing technologies(Joshi et al., 2024). Agricultural prediction tasks benefit from the high performance of algorithms which include Random Forest and Support Vector Machines and deep neural networks (Shawon et al., 2025). The combination of UAV imagery and satellite data provides researchers with tools for conducting thorough crop monitoring and environmental assessment which enhances yield prediction accuracy and enables data-driven precision farming implementation (Yuan et al., 2024)

Problem Statement

Machine learning-based crop yield prediction methods have achieved progress, yet they still face numerous existing challenges. Agricultural productivity is influenced by multiple interacting factors such as rainfall, temperature, soil conditions, and crop management practices, making prediction difficult.The majority of machine learning models require extensive high-quality datasets, which restrict their ability to perform effectively in various geographic areas that experience different weather patterns (Joshi et al., 2024) The advanced remote sensing techniques and UAV-based monitoring systems supply essential agricultural information, yet researchers still face obstacles when it comes to choosing useful features and understanding how their models function ((Yuan et al., 2024) The development of predictive models demands accurate results together with understandable evaluation methods which provide interpretable insights.

Current Industry Practice

Current precision agriculture systems use machine learning and satellite imaging and remote sensing technologies to track agricultural growth and predict harvest outcomes. Environmental and soil data analysis uses Random Forest and deep learning algorithms to achieve better prediction outcomes. UAV imagery and climate data enhance crop monitoring and yield prediction across different regions (Joshi et al., 2024; Yuan et al., 2024)

Aim of the Study

The study aims to create a machine learning system which uses environmental data and agricultural input data to predict crop yield while developing Explainable Artificial Intelligence methods to make its results more understandable.

Objectives of the Study

The study has three main objectives which researchers will achieve through their work:

- To examine how environmental variables like rainfall and temperature and weather patterns affect crop production.
- To assess how different agricultural practices such as fertilizer application and irrigation methods and soil characteristics impact agricultural output.
- To create machine learning regression models which will forecast crop yield.
- To assess how various machine learning algorithms perform in predicting crop yield.
- To use SHAP values to explain how their models make predictions through Explainable Artificial Intelligence techniques.

Research Questions

1. How do environmental factors (rainfall, temperature, weather conditions) impact crop yield in different regions, and what insights can be obtained using Explainable AI techniques?
2. What is the relative importance of agricultural inputs such as fertilizer, irrigation, and soil type in predicting crop yield as interpreted through SHAP values and feature importance?
3. Can machine learning regression models accurately predict crop yield, and which model performs best for this dataset? Additionally, how can Explainable AI enhance model interpretability for stakeholders such as farmers and policymakers?

Structure of the Report

The research topic of the study together with its objectives and research questions gets introduced in Chapter 1. The second chapter evaluates the existing research materials that connect to this study. Chapter 3 describes the dataset. The research paper explains ethical issues in Chapter 4. The research study describes its research methods in Chapter 5. The results of the study get presented in Chapter 6. The study presents its findings through analysis and discussion in Chapter 7. The study ends with Chapter 8 and Chapter 9 contains the reference materials.

Chapter 2: Background

Overview

This chapter reviews previous research related to crop yield prediction using machine learning techniques. The selected studies demonstrate how data-driven models enhance agricultural forecasting through their analysis of climate, soil, and crop-related variables. The four papers used in this chapter were selected because they are peer-reviewed journal articles that apply machine learning or deep learning techniques to real agricultural datasets and evaluate predictive performance using standard metrics. The research studies were selected because they use environmental and agricultural elements to improve crop yield prediction accuracy.

Literature Reviewed

Study 01

Researchers created a machine learning system which predicted crop yields while studying climate change effects on agricultural production. The dataset included information from 101 countries which the Food and Agriculture Organization and the World Bank collected between 1990 and 2013. The study evaluated multiple algorithms which included K-Nearest Neighbors and Decision Tree and Bagging Regressor and Random Forest and XGBoost. The researchers used an 80/20 train–test split to train the models while implementing preprocessing methods which included feature scaling and one-hot encoding. The research results demonstrated that the XGBoost ensemble model achieved its best performance through an R^2 score of 0.9745 (Islam et al., 2024)

Study 02

The study proposes a hybrid deep learning model for crop yield prediction by combining a 1D Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network, enhanced by an attention layer and skip connection. The model was tested on wheat and rice data sets to evaluate its performance against standard machine learning methods which included Support Vector Regressor, Decision Tree Regressor, and Random Forest Regressor. The proposed CNN-LSTM model achieved strong predictive performance with RMSE of 0.017, MAE of 0.09, and an R^2 score of 0.967, indicating high prediction accuracy (Kalmani et al., 2024).

Study 03

Another research study developed a machine learning system that predicts winter wheat yield through soil information, satellite images, and climate data. The research team used different spatial and temporal datasets to conduct their study while implementing feature selection methods to determine which variables had the greatest impact on crop yield. The researchers developed and tested machine learning models which included Logistic Regression, Random

Forest, and XGBoost to analyze multiple years of agricultural data. The study found that XGBoost outperformed both Logistic Regression and Random Forest because it delivered better predictive results and superior model performance. The research demonstrated that evapotranspiration, vegetation indices, and soil moisture variables serve as essential factors for yield prediction which requires environmental indicators to be included in forecasting models (Khosravani Shariati & Abbasi, 2025).

Study 04

Another study used machine learning models to study maize yield and nutrient efficiency in tropical agricultural regions. The researchers developed predictive models by using soil characteristics and fertilizer usage along with environmental variables as input features. The Random Forest algorithm proved to be the best method for forecasting maize yield and agronomic efficiency across various agro-ecological zones. The results showed that both soil variables and nitrogen application rates played essential roles in determining yield performance. The research results demonstrate that machine learning models can offer essential insights which help optimize fertilizer management and boost agricultural productivity in developing areas (Asamoah et al., 2024)

Gap identified

Previous studies show that machine learning and deep learning models improve crop yield prediction accuracy, but several limitations remain. The agricultural environment needs to be studied through various geographical regions because current research depends on datasets from specific locations. Research also focuses mainly on model performance rather than understanding interactions between environmental, temporal, and soil factors. The existing frameworks fail to combine different data types which include satellite imagery and weather data and soil information. The research community needs to create agricultural models which maintain their accuracy across multiple farming conditions.

Chapter 3: Dataset

The dataset used in this research was obtained from the Kaggle data repository. The dataset can be accessed by the public under the title "Agriculture Crop Yield Dataset[1]".

Samuel Oti Attakorah uploaded the dataset to Kaggle. Kaggle serves as a platform that machine learning and data science researchers use to access curated datasets which they can use for their experimentation and benchmarking and educational purposes.

Original Data Collection

Global agricultural organizations collect crop yield data through national agricultural departments and research institutions worldwide. The dataset contains historical agricultural data which researchers collected through surveys and farm reports and monitoring systems. The collected data serves three purposes which include analyzing crop production trends and monitoring agricultural productivity and supporting research about food security and climate impacts and sustainable farming practices.

Dataset Description

The dataset includes information about agricultural production across different countries and years. The main variables in the dataset include:

- Country/Area
- Year
- Crop Type
- Rainfall
- Temperature
- Pesticide Usage
- Yield

These variables allow researchers to analyze the relationship between environmental conditions and agricultural productivity.

Dataset Selection Justification

The dataset selection process required this dataset because it includes various environmental and agricultural elements which directly impact agricultural output. The dataset contains variables which include rainfall and temperature and pesticide usage and crop type and year information, so this dataset enables machine learning models to forecast agricultural output.

The dataset selection process required this dataset because it includes multi-year data which spans multiple countries, so researchers can study long-term agricultural patterns. The dataset

[1] https://www.kaggle.com/datasets/samuelotiattakorah/agriculture-crop-yield.

contains structured data which maintains a clean appearance, so it can be used to develop predictive models and test different machine learning algorithms.

Exploratory Data Analysis (EDA)

I conducted exploratory data analysis to investigate the dataset structure while searching for missing values and outlier detection.

The first step in EDA required researchers to investigate the dataset dimensions and attribute details and the types of data present. The dataset contains several thousand records representing crop production data across multiple countries and years. After that I created data distribution graph which are as follows

Boxplot

The yield distribution among cotton rice barley soybean wheat and maize crops appears in Figure 3.1. The median yield of the crop produces 4.5 tons per hectare while its yield ranges between -1 ton and 10 tons per hectare.

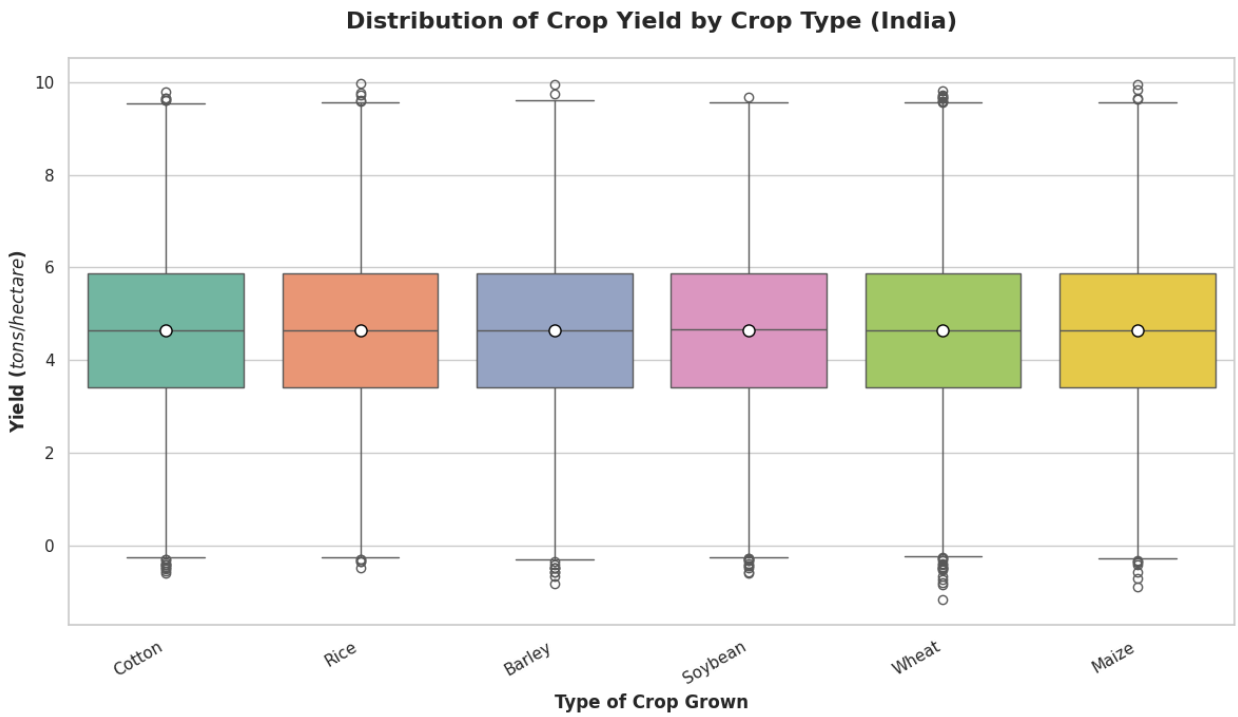

Figure 3.1: Boxplot showing the distribution of crop yield

Distribution of annual Rainfall

Figure 3.2 shows the annual rainfall distribution which shows rainfall patterns throughout the studied area. The study collected rainfall data which showed a range from 100 millimeters to 1000 millimeters with most measurements falling between 300 millimeters and 900 millimeters.

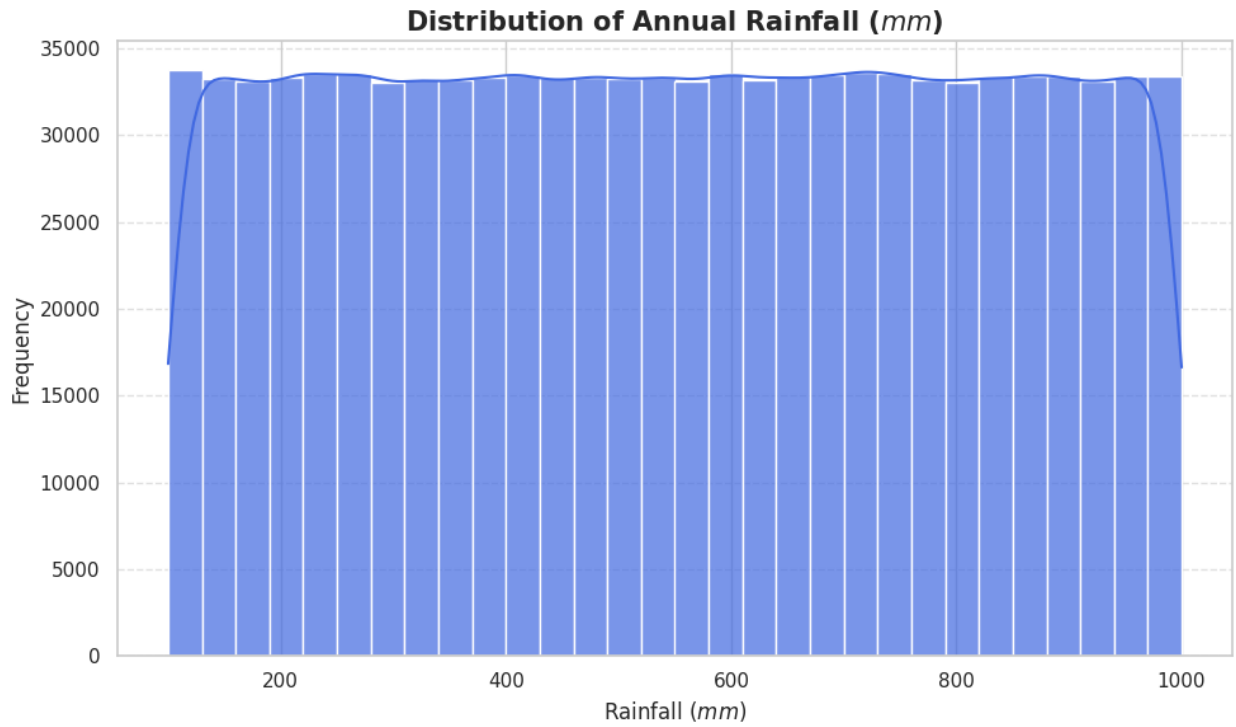

Figure 3.2: Histogram displaying the distribution of annual rainfall values recorded in the dataset.

Distribution of Temperature

The study presents temperature distribution data which shows how temperatures vary throughout the research area. The agricultural temperature data shows values that range from 15 degrees Celsius to 40 degrees Celsius with most agricultural data points existing between 20 degrees Celsius and 35 degrees Celsius.

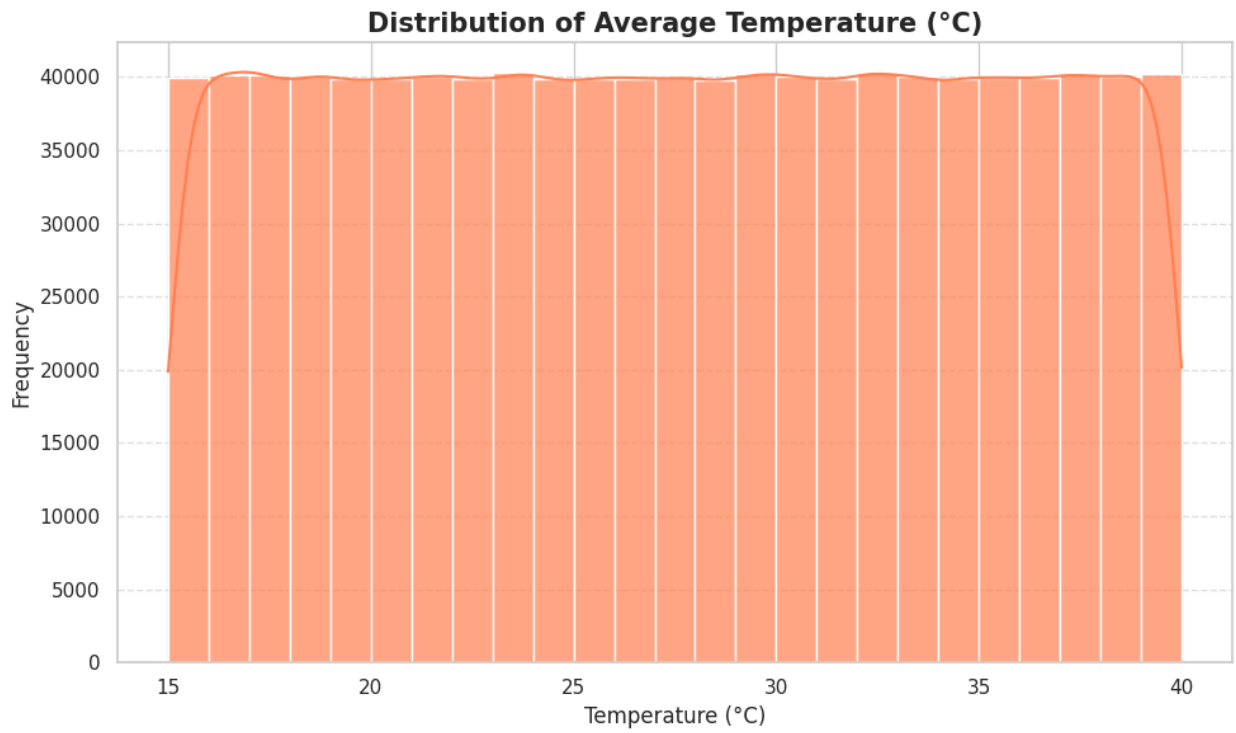

Figure 3.3: Histogram showing the distribution of average temperature values affecting crop growth.

Distribution of Crop Duration

Figure 3.4 shows crop growth duration ranging from 60 to 150 days. Most crops are harvested between 80 and 130 days, indicating moderate variation across different crop types.

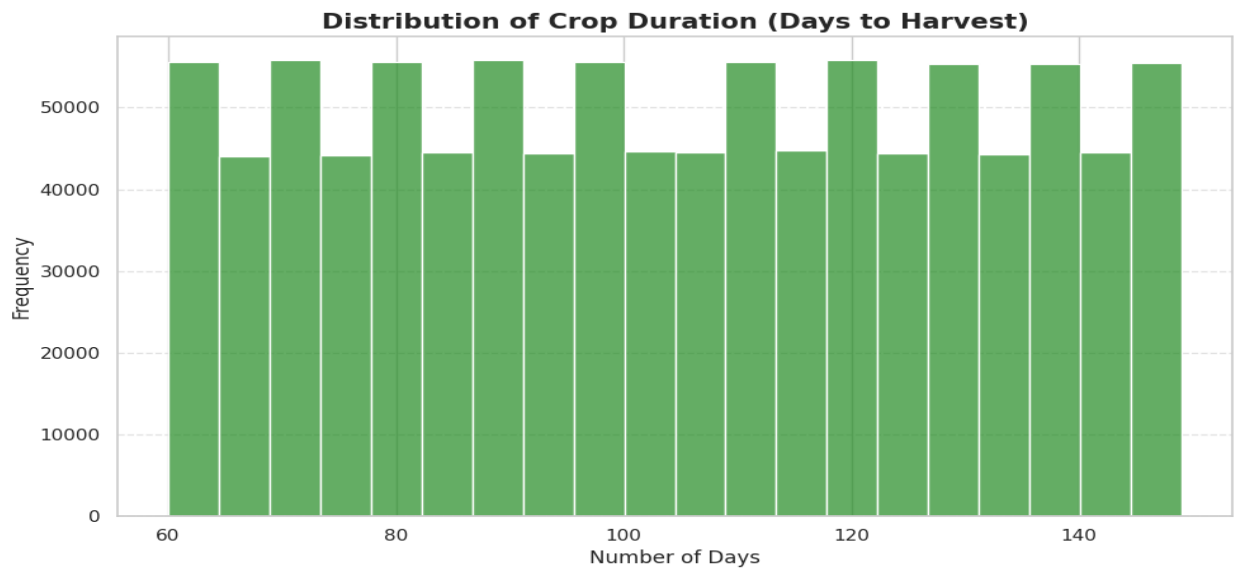

Figure 3.4: Histogram illustrating the distribution of crop duration in days from planting to harvest.

Overall Distribution of Crop Yield

Figure 3.5 presents the overall yield distribution. Crop yield ranges approximately from 0 to 10 tons/hectare, with most observations concentrated between 3 and 7 tons/hectare.

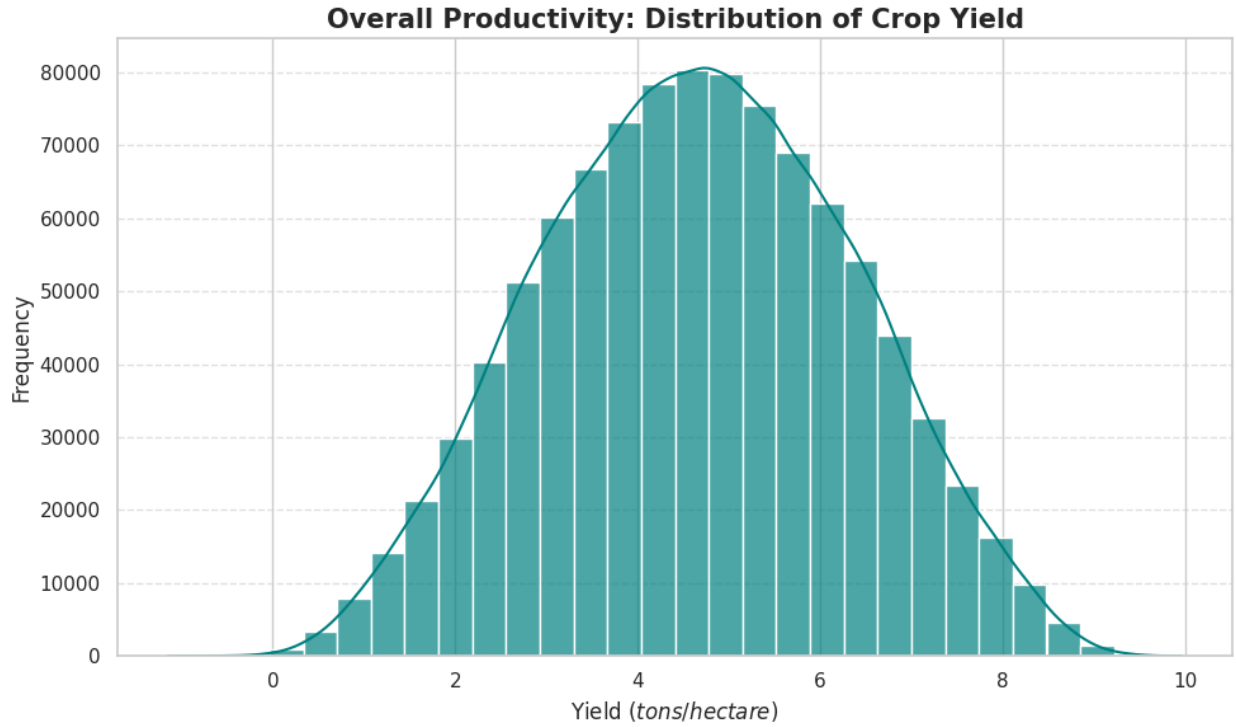

Figure 3.5: Histogram showing the overall distribution of crop yield values in tons per hectare across the dataset.

Data Preprocessing

The dataset required several preprocessing steps to achieve proper preparation. First, negative yield values were removed because crop yield cannot be less than zero. Next, Boolean variables (Fertilizer_Used and Irrigation_Used) were converted from True/False to 1 and 0.

Label Encoding conversion was used to change categorical features (Region, Soil_Type, Crop, Weather_Condition) into their respective numerical values. The Country column was removed since it contained only one value and had no predictive importance.

The StandardScaler function was used to normalize all input features of the data except for the target variable (Yield_tons_per_hectare) which improved the model performance.

Chapter 4: Ethical Consideration

Data Source and Permission

Your training data extends through the month of October in the year 2023. The dataset used in this study was obtained from the Kaggle platform. It is distributed under the Creative Commons CC0 Public Domain license which allows researchers to use the material without any fees while they work on their studies.

Personal Data and GDPR

The dataset contains agricultural data which includes crop type and rainfall and temperature and yield data. The dataset does not contain any personal data which makes it exempt from GDPR requirements.

Ethical Approval

The study does not involve human participants nor does it include surveys or individual data collection activities. The research does not require ethical approval because it does not involve any human participants.

Ethical Data Collection

The dataset was uploaded by Samuel Oti Attakorah and it includes non-sensitive agricultural data which researchers will use for their machine learning studies.

Chapter 5: Methodology

Chapter Overview

In this chapter I explains the methodology used in my research to develop machine learning models for predicting crop yield. The chapter describes my dataset preparation process and model selection process and model training process and parameter optimization process and performance evaluation process. The methodology developed predictive models to analyze agricultural and environmental elements and forecast agricultural production.

Libraries and Tools Used

My research used Python programming language together with various data science libraries to conduct data analysis and machine learning operations. The Pandas and NumPy libraries were used for data manipulation, numerical computation, and dataset management. I used Matplotlib and Seaborn to create graphs and plots for data visualization because they help demonstrate data patterns.

Data Preparation

I used the Scikit-learn library for machine learning because it provided tools needed to train and evaluate models and process data, while SHAP library served as the tool to explain model outputs. The dataset was divided into input features and the target variable (Yield_tons_per_hectare). The model training used 80% of the data while 20% was reserved to test the model performance.

Model Training

I trained three regression models which predict crop yield throughout this study. The models include linear modeling and decision tree learning and ensemble learning as different machine learning methods. Table 5.1 shows the basic model configuration used in my study

Table 5.1: Basic Model Configuration

Model	Key Parameters	Configuration
Linear Regression	Default parameters	Baseline regression model
Decision Tree Regressor	max_depth = 10	Controls tree complexity to reduce overfitting
Random Forest Regressor	n_estimators = 50, max_depth = 10	Ensemble model combining multiple decision trees

Description and Technical Rationale

Linear Regression

I used Linear Regression as a baseline model to determine whether crop yield could be predicted using simple linear relationships between environmental variables and yield.

Decision Tree Regressor

Decision Tree is a non-linear model that splits the dataset into smaller subsets based on feature values. I used this model because agricultural data often contains complex relationships between variables such as rainfall, soil type, and temperature.

Random Forest Regressor

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. I selected this model because ensemble methods usually perform well on datasets with nonlinear relationships and multiple interacting variables.

Hyperparameter Optimization

The GridSearchCV and RandomizedSearchCV methods from Scikit-learn library served as my tools for performing hyperparameter optimization because these methods helped me achieve better model performance results. The techniques evaluate different combinations of parameters until they identify the optimal settings which work best with each specific model. Table 5.2 shows the search space used in Hyperparameter Optimization.

Table 5.2: Optimization Search Space

Model	Parameter	Search Values
Ridge Regression	alpha	0.1, 1.0, 10.0
Decision Tree	max_depth	10, 20
	min_samples_leaf	5, 10
Random Forest	n_estimators	50
	max_depth	10, 20
	min_samples_split	5

The optimization process required me to use 100,000 records from the dataset because this sample size provided enough data to maintain the original data characteristics. The optimized models underwent training with complete training data after we discovered the optimal model parameters.

Model Evaluation

The testing dataset served as my basis for testing the performance of the trained models after I finished the optimization procedure. The prediction accuracy assessment used multiple regression metrics to evaluate results.

- The Mean Absolute Error (MAE) calculates the average discrepancy that exists between estimated crop yield values and actual crop yield values.

- Mean Squared Error (MSE) determines the squared difference between predicted values and actual values which results in higher penalties for prediction errors that exceed a specific threshold.
- The Root Mean Squared Error (RMSE) displays prediction errors through the same measurement system that applies to crop yield data.
- The R^2 Score (Coefficient of Determination) shows the extent to which a model can account for variations in crop yield data.

K-Fold Cross Validation

The evaluation of model stability and reliability testing was performed through the application of 5-Fold Cross Validation method. The dataset was divided into five folds where four folds were used for training and one for validation. The process conducted five repetitions which permitted each fold to function as the validation component. The final performance assessment combined all folds to calculate an average which decreased bias while increasing assessment dependability.

Model Interpretability

I performed feature importance analysis through Random Forest model testing to investigate how machine learning models create their predictions. The analysis shows which environmental and agricultural factors most effectively predict crop yield outcomes.

I applied the SHAP (SHapley Additive Explanations) method to evaluate the impact of each feature on prediction results. SHAP explains how different variables including rainfall and irrigation and fertilizer usage and soil type and temperature affect predicted crop yields. The research demonstrates model performance through better transparency which helps users understand machine learning model behavior.

Chapter 6: Results

Overview

The results of my research study on crop yield prediction through machine learning models which I developed are presented in this chapter. The study assessed model accuracy through multiple regression metrics which provided reliable measurement results. The task required regression analysis which made classification metrics like accuracy and confusion matrices unsuitable for evaluation. The evaluation used Mean Absolute Error (MAE) and Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) and R^2 Score. The metrics assess prediction errors and explained variance which enables full assessment of model performance and prediction accuracy.

Model Performance Before Hyperparameter Optimization

The initial performance of the models before hyperparameter optimization is presented in Table 6.1.

Table 6.1: Model Performance Before Optimization

Model	MAE	RMSE	R^2 Score
Linear Regression	0.3983	0.4993	0.9132
Decision Tree	0.4008	0.5022	0.9122
Random Forest	0.3991	0.5002	0.9129

The three models reached excellent predictive accuracy because their R^2 scores exceeded 0.91 according to the results presented in Table 6.1. The highest R^2 score of 0.9132 and lowest RMSE value for Linear Regression demonstrated that this model used data for better predictive performance than its competitors. Random Forest produced similar results, while Decision Tree showed marginally higher prediction error. Figure 6.1 shows the graph for model comparison

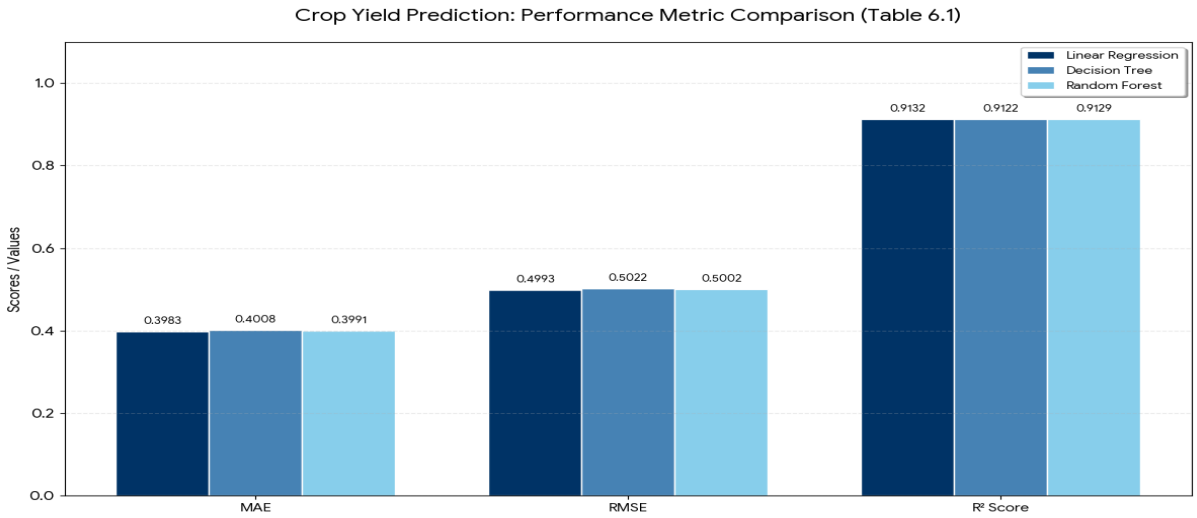

Figure 6.1: Model Performance

Model Performance After Hyperparameter Optimization

I used hyperparameter optimization to enhance the performance of their models. The optimized model results are shown in Table 6.2.

Table 6.2: Model Performance After Optimization

Model	R^2 Score	MSE	RMSE
Ridge Regression	0.9132	0.2493	0.4993
Decision Tree	0.9122	0.2522	0.5022
Random Forest	0.9129	0.2502	0.5002

The optimization process did not improve model performance because baseline models already had correct settings. The R^2 value of Ridge Regression proved that linear relationships between variables serve as key factors for predicting crop yield. Random Forest achieved strong results because it could identify nonlinear patterns in data.

K-Fold Cross-Validation Results

I used 5-Fold Cross-Validation to test model stability. The results are shown in Table 6.3.

Table 6.3: K-Fold Cross-Validation Results

Model	Average R^2	Average RMSE
Linear Regression	0.9130	0.5001
Decision Tree	0.9119	0.5032
Random Forest	0.9126	0.5011

The models achieved consistent results throughout the dataset testing which included multiple dataset testing sessions. The average R^2 values remain around 0.91 which shows that the models perform well in real-world situations while they display only minor overfitting issues.

Explainable AI Results

Feature Importance Analysis

The feature importance analysis (see Figure 6.2) identifies the variables that contribute most to crop yield prediction. The results show that rainfall is the most influential factor affecting crop productivity. Crop yield results depend on two agricultural inputs which include fertilizer usage and irrigation practices.

SHAP Analysis

The SHAP summary plot (see Figure 6.2) shows how individual variables influence the prediction output. The combination of increased rainfall and higher fertilizer application leads to higher predicted crop yield values while decreased rainfall results in lower predicted productivity.

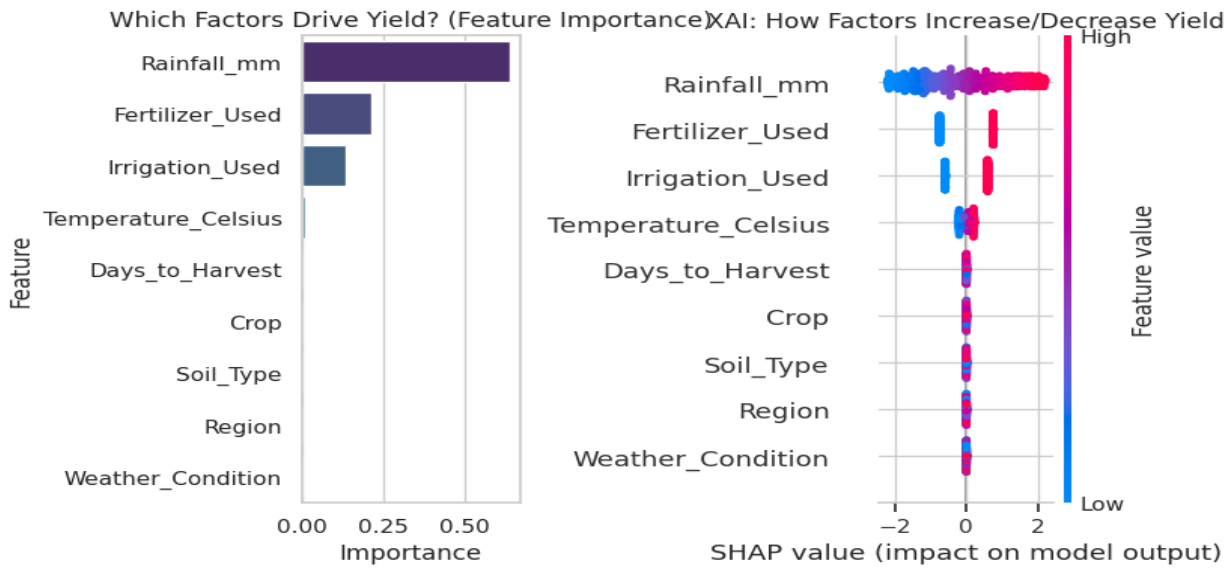

Figure 6.2: Model Interpretability

Relative Importance

Figure 6.3 displays how essential agricultural inputs impact crop yield estimation. The bar chart shows that fertilizer application provides the highest impact to the model, which receives a score of approximately 0.21, while irrigation follows with a score of approximately 0.14 and soil type shows minimal impact. The SHAP dependence plot shows that fertilizer is applied when value equals 1 because the SHAP values reach positive values between 0.75. When fertilizer is not used (value = -1) the SHAP values become negative at approximately -0.75 which shows lower expected yield results.

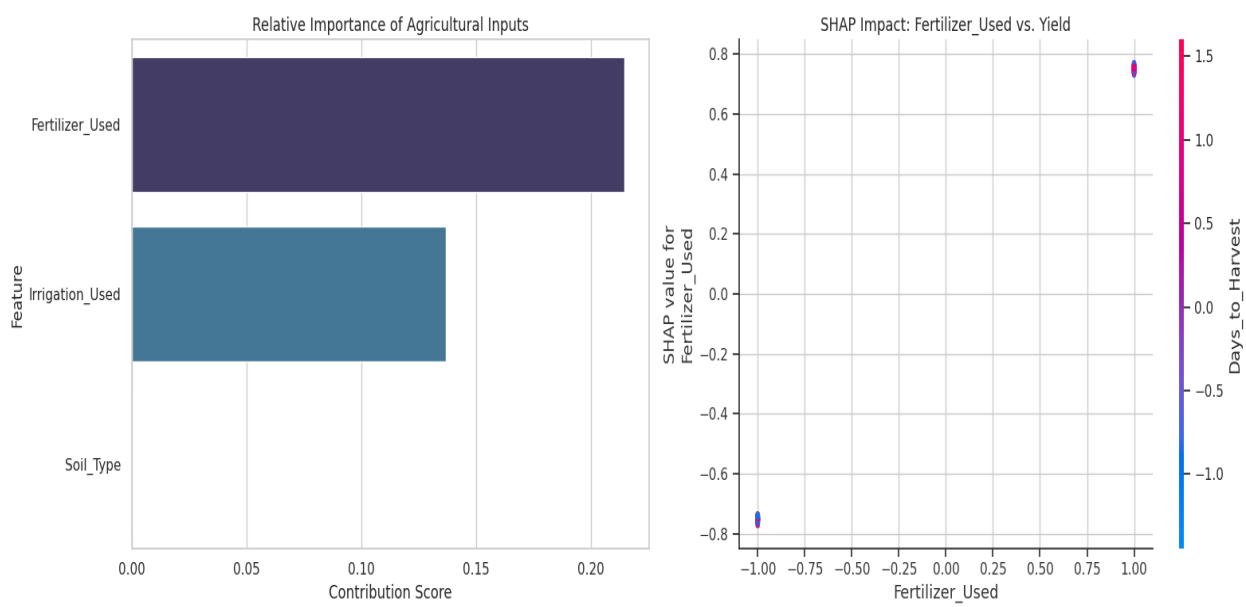

Figure 6.3: Relative importance of agricultural inputs and SHAP analysis

Research Questions Addressed

The research questions receive their answer through the results which demonstrate that environmental factors especially rainfall impact crop yield predictions. The application of agricultural inputs through fertilizer and irrigation methods helps to increase crop yield results. The machine learning models reached their highest prediction accuracy at $R^2 \approx 0.91$ while Explainable AI methods revealed the prediction impact of various variables.

Practical Implications

The findings of this study demonstrate that machine learning models can support data-driven agricultural decision-making. Predictive models enable farmers to forecast crop yield by using environmental data and their farming methods.

The Explainable AI techniques create model transparency through their ability to show how various factors influence yield predictions. This transparency enables farmers and agricultural planners and policymakers to make educated choices regarding irrigation methods and fertilizer application and crop management strategies.

The results show that machine learning with explainable AI provides essential knowledge which helps enhance agricultural productivity and enables sustainable agricultural methods.

Chapter 7: Analysis and Discussion

Interpretation of Results

The research results show that machine learning models can successfully predict crop yield through the analysis of environmental and agricultural data. The three regression models achieved strong predictive accuracy because their R^2 values approached 0.91. This indicates that the models were able to explain approximately 91% of the variation in crop yield within the dataset. The low RMSE values (around 0.50) function as evidence that prediction errors remained within minimal limits. The results indicate that the environmental and agricultural variables present in the dataset effectively predict crop yield through their valuable information content.

The results also show that environmental variables such as rainfall play a major role in crop yield prediction. The feature importance analysis discovered that rainfall served as the primary environmental factor which most affected crop productivity. This finding aligns with agricultural knowledge because rainfall directly affects soil moisture, plant growth, and crop development. The research showed that agricultural management practices including fertilizer application and irrigation directly impacted the resulting agricultural yield.

Best Performing Model

Linear Regression achieved better results than Decision Tree and Random Forest models because it displayed superior R^2 score and RMSE performance. The dataset shows that crop yield depends on input variables through a mostly linear connection. Linear Regression performed well because it models direct relationships between predictors and the target variable. The linear model successfully captured crop yield patterns because rainfall and fertilizer usage and irrigation serve as consistent predictors of crop yield.

Random Forest achieved strong performance through its ability to detect both nonlinear patterns and hidden connections among different factors. Random Forest uses ensemble methods which merge multiple decision trees to enhance prediction accuracy while decreasing overfitting risks. The dataset showed only minor improvement over Linear Regression because it contained no complex nonlinear patterns which required advanced modeling techniques.

The Decision Tree model delivered slightly lower results than the other tested models. Decision trees can easily overfit the training data when they become too complex. The model developed through hyperparameter controlled tree depth which led to the acquisition of data noise that damaged its ability to generalize results.

Comparison with Literature

The study results display complete agreement with existing research findings. Previous studies such as (Islam et al., 2024) and (Khosravani Shariati & Abbasi, 2025).showed that machine learning models can successfully predict crop yield using environmental and agricultural variables. The studies achieved excellent predictive results through ensemble learning techniques that included XGBoost and Random Forest which produced high R^2 scores.

The research conducted (Kalmani et al., 2024) showed that deep learning models could identify intricate agricultural patterns when researchers utilized extensive datasets and various data sources. The process of deep learning model development requires both extensive datasets and substantial computational power. The research shows that basic machine learning techniques like Linear Regression and Random Forest can produce accurate predictions when agricultural data with important features exists in the dataset.

Research demonstrates that environmental factors which include rainfall and soil moisture and climate indicators play a crucial role in determining crop yield (Asamoah et al., 2024). The study results about feature importance showed that rainfall and fertilizer usage and irrigation serve as the primary factors which determine crop yield.

Limitations of the Study

The research demonstrates multiple limitations which exist despite the models achieving accurate predictive results. The study utilized a dataset which included simulated and aggregated agricultural data but did not contain actual agricultural monitoring data from farms. The dataset used in this research fails to capture all additional factors which affect real-world agricultural operations including pest outbreaks and soil nutrient fluctuations and farming practices.

The dataset has a geographic scope which limits its applications. Agricultural productivity can vary significantly across different regions due to differences in climate, soil characteristics, and farming practices. The results of one dataset study do not permit researchers to make accurate predictions about agricultural outcomes which will occur in different agricultural environments.

The models use structured numerical data elements as their foundational components. The incorporation of satellite imagery and remote sensing data and seasonal climate indicators as additional data sources has the potential to enhance forecasting precision.

Relationship to Research Objectives

The study results provide direct answers to the research objectives. The first objective was to examine how environmental variables influence crop yield. Rainfall functions as an essential environmental factor which determines crop productivity according to the results of feature importance and SHAP analysis.

The second objective was to analyze the role of agricultural inputs such as fertilizer and irrigation. The results clearly show that fertilizer usage and irrigation significantly contribute to yield prediction

The third objective was to develop machine learning models capable of predicting crop yield. The results confirm that machine learning regression models can achieve high prediction accuracy for this dataset.

Practical Applications

The agricultural decision-support systems need practical agricultural decision-support systems which can use the models developed in this research. The predictive models which describe environmental conditions and farming practices will enable farmers and agricultural planners to estimate their crop yield. The system enables farmers to optimize their irrigation methods and fertilizer use and crop planning activities.

XAI techniques use SHAP as an Explainable Artificial Intelligence method which helps users understand model output through its transparent prediction mechanism. Farmers and policymakers can understand how specific factors influence predicted yield rather than relying on a black-box model.

Addressing the Research Questions

The experimental results successfully solved the research questions which needed to be solved. The combination of rainfall and agricultural inputs which include fertilizer and irrigation systems creates significant effects on crop yield prediction models. I achieved excellent predictive results through their machine learning regression models while explainable AI methods showed how various factors impacted model output.

Chapter 8: Conclusion

This research used environmental factors together with agricultural data to build machine learning models which predicted crop yield. The results showed that all three regression models achieved strong predictive performance with R^2 values close to 0.91, indicating reliable yield prediction. The results showed that Linear Regression produced better results than other models while Random Forest showed consistent performance. The analysis used Feature importance and SHAP methods to identify rainfall and fertilizer usage together with irrigation as the primary factors which determined crop yield. The study results show that machine learning together with explainable AI enables data-based agricultural decision support.

The proposed approach enables farmers and agricultural planners to predict crop yield while optimizing their resource allocation processes. Future research could use satellite images together with current weather information and extensive regional datasets to improve prediction capabilities and create models that apply to various farming situations.

Chapter 9: References

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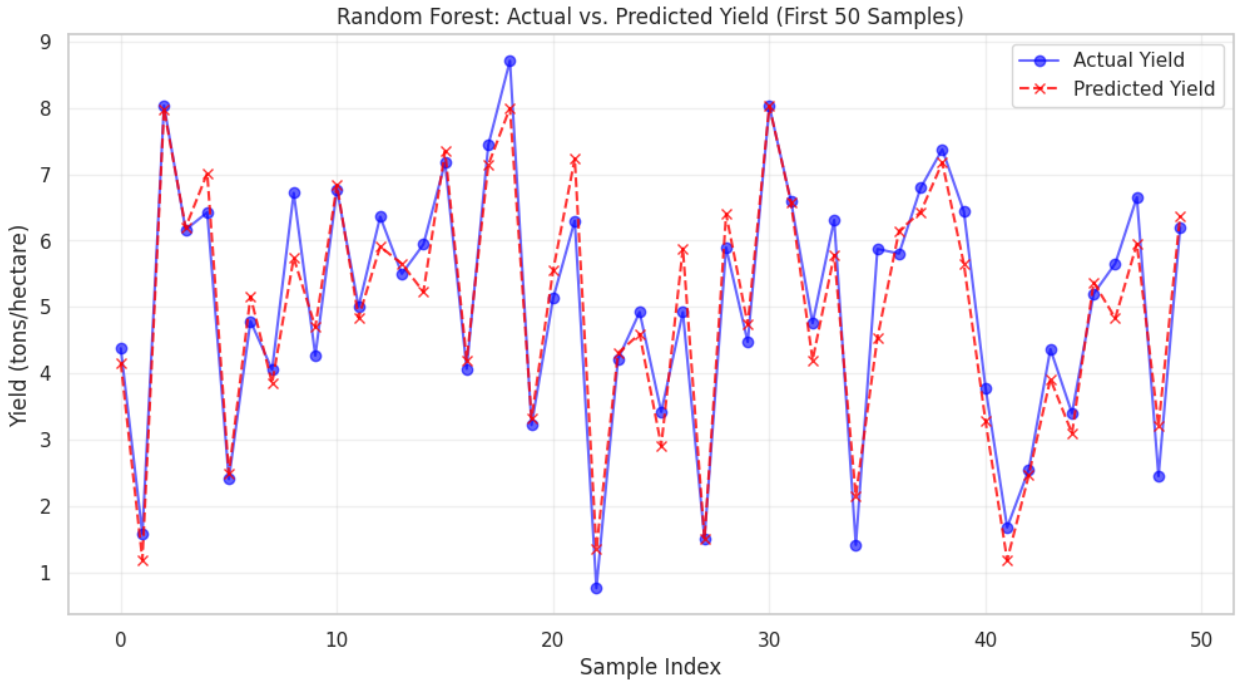
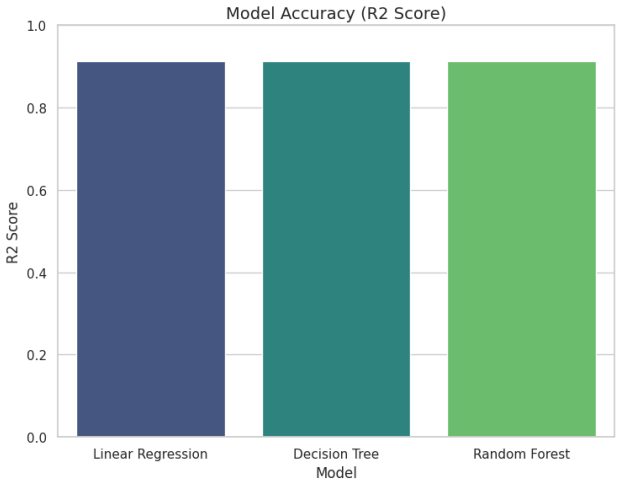
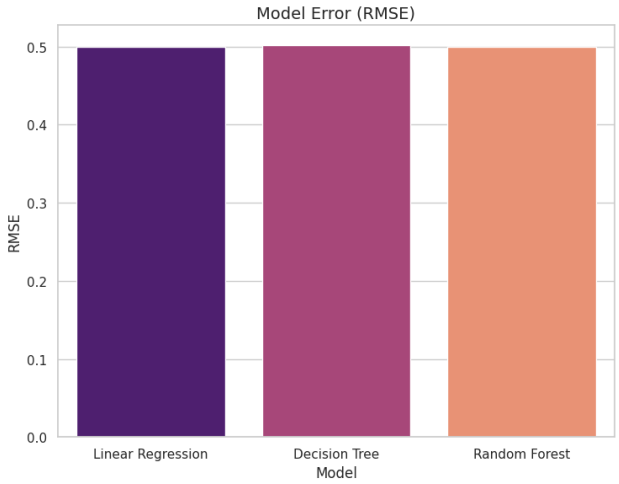
Appendix:


Random Forest: Actual vs. Predicted Yield (First 50 Samples)
Actual Yield
Predicted Yield
Yield (tons/hectare)
Sample Index
0
10
20
30
40
50
1
2
3
4
5
6
7
8
9


Model Accuracy (R2 Score)
R2 Score
0.0
0.2
0.4
0.6
0.8
1.0
Linear Regression
Decision Tree
Random Forest
Model


Model Error (RMSE)
RMSE
0.0
0.1
0.2
0.3
0.4
0.5
Linear Regression
Decision Tree
Random Forest
Model